

Problem Set 2

Applied Stats/Quant Methods

Matilda Tomatis

Due: October 23, 2025

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday October 23, 2025. No late assignments will be accepted.

Question 1: Political Science

The following table was created using the data from a study run in a major Latin American city.¹ As part of the experimental treatment in the study, one employee of the research team was chosen to make illegal left turns across traffic to draw the attention of the police officers on shift. Two employee drivers were upper class, two were lower class drivers, and the identity of the driver was randomly assigned per encounter. The researchers were interested in whether officers were more or less likely to solicit a bribe from drivers depending on their class (officers use phrases like, “We can solve this the easy way” to draw a bribe). The table below shows the resulting data.

¹Fried, Lagunes, and Venkataramani (2010). “Corruption and Inequality at the Crossroad: A Multi-method Study of Bribery and Discrimination in Latin America. *Latin American Research Review*. 45 (1): 76-97.

	Not Stopped	Bribe requested	Stopped/given warning
Upper class	14	6	7
Lower class	7	7	1

- (a) Calculate the χ^2 test statistic by hand/manually (even better if you can do "by hand" in R).
- (b) Now calculate the p-value from the test statistic you just created (in R).² What do you conclude if $\alpha = 0.1$?
- (c) Calculate the standardized residuals for each cell and put them in the table below.

	Not Stopped	Bribe requested	Stopped/given warning
Upper class			
Lower class			

- (d) How might the standardized residuals help you interpret the results?

Answer a

In order to manually compute the χ^2 test statistic we must go through various computational steps. Firstly, we must compute the total sum of each row and of each column, as well as the grand total of observations for the whole table. These values will be necessary for the calculation of expected frequencies.

²Remember frequency should be > 5 for all cells, but let's calculate the p-value here anyway.

A quick way to do so is by "pivoting longer" the dataset. With this operation the dataset will be tidy (i.e. there would be a variable for each column, a requirement not respected let the dataset be left wide) and from there computing the sum of observations grouping by "Class" and by "Observed outcome".

After obtaining these necessary values, we can proceed to calculate the expected frequencies for each observation via the formula $\frac{\text{Row Total} \times \text{Column Total}}{\text{Grand Total}}$.

```
1 polsci_df_long <- polsci_df_long |>
2 mutate(Fe = (Rows_tot * Cols_tot) / GT)
```

At this point every necessary element has been obtained and the χ^2 test statistic can be manually determined. This is the reference formula for the computation: $\chi^2 = \sum \frac{(F_o - F_e)^2}{F_e}$

```
1 Chi_squared <- sum(((polsci_df_long$Fo - polsci_df_long$Fe)^2) / polsci_df_long$Fe)
```

```
[1] "The Chi-squared value found is: 3.7912"
```

Answer b

Once the test statistic is obtained we can calculate the related p-value.

To do so we must consider a chi-squared distribution with " (number of rows-1) × (number of columns-1) " degrees of freedom.

```
1 df <- (2-1)*(3-1)
2 p_value <- pchisq(Chi_squared, df, lower.tail = FALSE)
```

```
[1] "p-value: 0.1502 -
```

```
The variables don't showcase statistically significant association"
```

Considering the p-value obtained is bigger than the significance level α , we cannot reject our null hypothesis of statistical independence between the variables "Social Class of the driver" and "Observed bribery request".

Paraphrasing our find, the variables under scrutiny appear to be statistically independent.

Answer c

We must now calculate the standardized residuals for each cell of our data frame. The

formula for the computation is the following: $z = \frac{F_o - F_e}{\sqrt{F_e \left(1 - \frac{R_{\text{tot}}}{G_T}\right) \left(1 - \frac{C_{\text{tot}}}{G_T}\right)}}$.

```

1 polsci_df_long <- polsci_df_long |>
2   mutate(z_values = ((Fo - Fe) /
3     sqrt((Fe)*(1 - Rows_tot / GT) * (1 - Cols_tot / GT))))

```

For the displaying of the results the tibble will be "pivoted wider" once again.

Table 1: Standardized Residuals for Class and Police Stop Outcomes

	Class	Not stopped	Bribe requested	Stopped/given warning
1	Upper class	0.322	-1.642	1.523
2	Lower class	-0.322	1.642	-1.523

Answer d

Standardized residuals are useful for the interpretation of a chi-squared coefficient as they indicate the direction and magnitude of divergence of the observed values from the expected ones. The chi-squared coefficient alone does not allow for any commentary on such informations. In principle, it might be inflated by a few larger deviations rather than by the whole sample and it would be complicated to determine simply by its value. In our case, we can observe how none of the z-values exceed ± 2 , a general rule of thumb threshold beyond which it's plausible to question the independence of the variables under examination.

Question 2: Economics

Chattopadhyay and Duflo were interested in whether women promote different policies than men.³ Answering this question with observational data is pretty difficult due to potential confounding problems (e.g. the districts that choose female politicians are likely to systematically differ in other aspects too). Hence, they exploit a randomized policy experiment in India, where since the mid-1990s, $\frac{1}{3}$ of village council heads have been randomly reserved for women. A subset of the data from West Bengal can be found at the following link: <https://raw.githubusercontent.com/kosukeimai/qss/master/PREDICTION/women.csv>

Each observation in the data set represents a village and there are two villages associated with one GP (i.e. a level of government is called "GP"). Figure ?? below shows the names and descriptions of the variables in the dataset. The authors hypothesize that female politicians are more likely to support policies female voters want. Researchers found that more women complain about the quality of drinking water than men. You need to estimate the effect of the reservation policy on the number of new or repaired drinking water facilities in the villages.

Table 2: Names and description of variables from Chattopadhyay and Duflo (2004).

Name	Description
GP	An identifier for the Gram Panchayat (GP).
village	Identifier for each village.
reserved	Binary variable indicating whether the GP was reserved for women leaders or not.
female	Binary variable indicating whether the GP had a female leader or not.
irrigation	Number of new or repaired irrigation facilities in the village since the reserve policy started.
water	Number of new or repaired drinking-water facilities in the village since the reserve policy started.

³Chattopadhyay and Duflo. (2004). "Women as Policy Makers: Evidence from a Randomized Policy Experiment in India. *Econometrica*. 72 (5), 1409-1443.

- (a) State a null and alternative (two-tailed) hypothesis.
- (b) Run a bivariate regression to test this hypothesis in R (include your code!).
- (c) Interpret the coefficient estimate for reservation policy.

Answer a

1. H_0 states: $\beta_1 = 0$, interpretable as: having reserved positions for female leaders has no impact on the creation and repair of irrigation facilities.
2. H_a states: $\beta_1 \neq 0$, hence part of the leading positions being reserved for women does have an impact on the creation and repair of irrigation facilities.

Answer b

We will proceed to test our hypothesis by running the following bivariate regression:

$$\text{Irrigation} = \beta_0 + \beta_1 \text{Reserved} + \varepsilon$$

```
1 irrigation_lm <- lm( irrigation ~ reserved , data = women)
```

Answer c

The β_1 estimate has a value of -0.369, indicating that when a Gram Panchayat (GP) has positions reserved for female leaders the number of new or repaired irrigation facilities in its villages decreases by approximately one-third, on average, compared to non-reserved GPs. At all conventional significance levels (0.10, 0.05, and 0.01), the estimated coefficient indicates that reserving leadership positions for women in Gram Panchayats has no statistically significant effect on the creation or restoration of irrigation facilities. These results are in line with H_0 .

However, the model might suffer from oversimplification. It is difficult to hypothesize that only the reservation policy has an impact on irrigation facility funding: our estimate might suffer from omitted variable bias.

Table 3: Bivariate regression model with GP as Y and irrigation as X1

	<i>Dependent variable:</i>
	irrigation
reserved	−0.369 (1.122)
Constant	3.388*** (0.650)
Observations	322
R ²	0.0003
Adjusted R ²	−0.003
Residual Std. Error	9.506 (df = 320)
F Statistic	0.108 (df = 1; 320)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01